

Exercise 8

AIM :

To write a C program for the below requirements

1. Library management system -- Using structure to store the details of the books like Name, Price and ID of the book. create functions to upload details of 5 quantity of books and create a function to print the information of the book based on the ID

ALGORITHM :

1. START
2. DEFINE: MAX_BOOKS = 5, MAX_NAME = 100
3. DEFINE STRUCTURE Book with fields: name[], price, id
4. MAIN FUNCTION:
 - a. DECLARE: library[MAX_BOOKS], choice, searchID
 - b. DISPLAY program header
 - c. CALL uploadBooks(library, MAX_BOOKS)
 - d. CALL displayAllBooks(library, MAX_BOOKS)
 - e. DO:
 - i. DISPLAY menu with 3 options
 - ii. READ user choice
 - iii. SWITCH(choice):
 - CASE 1: READ searchID, CALL printBookByID()
 - CASE 2: CALL displayAllBooks()
 - CASE 3: DISPLAY exit message
 - DEFAULT: DISPLAY error message
 - f. WHILE choice != 3
 - g. RETURN 0
5. UPLOAD BOOKS FUNCTION:
 - a. FOR i = 0 to size-1:

- i. DISPLAY "Book i+1 details"
- ii. READ book ID
- iii. CLEAR input buffer
- iv. READ book name
- v. REMOVE newline from name
- vi. READ book price
- b. DISPLAY success message

6. PRINT BOOK BY ID FUNCTION:

- a. SET found = 0
- b. FOR i = 0 to size-1:
 - i. IF books[i].id == searchID:
 - DISPLAY book details
 - SET found = 1
 - BREAK loop
- c. IF found == 0: DISPLAY "Book not found"

7. DISPLAY ALL BOOKS FUNCTION:

- a. IF size == 0: DISPLAY "No books", RETURN
- b. DISPLAY table header
- c. FOR i = 0 to size-1:
 - i. DISPLAY formatted book details
- d. DISPLAY table end

8. END

OUTPUT :

```
PS D:\C programming\DHAVAMANI_EX8> gcc main.c -o strct
PS D:\C programming\DHAVAMANI_EX8> ./strct.exe

-----LIBRARY MANAGEMENT SYSTEM-----

=== UPLOAD BOOK DETAILS ===

--- Enter details for Book 1 ---
Enter Book ID: 1
Enter Book Name: harry potter and the sorcerer's stone
Enter Book Price: 299

--- Enter details for Book 2 ---
Enter Book ID: 2
Enter Book Name: harry potter and the chamber of secrets
Enter Book Price: 299

--- Enter details for Book 3 ---
Enter Book ID: 3
Enter Book Name: harry potter and the prisoner of azkaban
Enter Book Price: 299

--- Enter details for Book 4 ---
Enter Book ID: 4
Enter Book Name: harry potter and the goblet of fire
Enter Book Price: 299

--- Enter details for Book 5 ---
Enter Book ID: 5
Enter Book Name: harry potter and the order of the phoenix
Enter Book Price: 299

5 books uploaded successfully!
```

ALL BOOKS IN LIBRARY

ID	NAME	PRICE
1	harry potter and the sorcerer's stone	\$ 299.00
2	harry potter and the chamber of secrets	\$ 299.00
3	harry potter and the prisoner of azkaban	\$ 299.00
4	harry potter and the goblet of fire	\$ 299.00
5	harry potter and the order of the phoenix	\$ 299.00

----SEARCH MENU----

1. Search book by ID
2. Display all books
3. Exit

Enter your choice (1-3): 1

Enter Book ID to search: 2

=== BOOK FOUND ===

Book ID: 2

Book Name: harry potter and the chamber of secrets

Book Price: \$299.00

----SEARCH MENU----

1. Search book by ID
2. Display all books
3. Exit

Enter your choice (1-3): 2

ALL BOOKS IN LIBRARY

ID	NAME	PRICE
1	harry potter and the sorcerer's stone	\$ 299.00
2	harry potter and the chamber of secrets	\$ 299.00
3	harry potter and the prisoner of azkaban	\$ 299.00
4	harry potter and the goblet of fire	\$ 299.00
5	harry potter and the order of the phoenix	\$ 299.00

----SEARCH MENU----

1. Search book by ID
2. Display all books
3. Exit

Enter your choice (1-3): 3

Thank you for using Library Management System!

Exiting...